Geospatial proposal assessment

USD describes a 3D model's shape and size. Geospatial information says where it belongs on Earth, so models, surveys and measurements can work together.

Proposal quality means that another implementer can derive the inputs, evaluation order and observable outcomes without guessing. The full draft specifies those rules for the workflows shown here.

Selected demonstrations support the proposed behavior. Detailed conventions still need group review, and two prototype limitations remain. This evidence does not establish exhaustive completeness or independent geodetic accuracy.

A geographic location for a 3D model

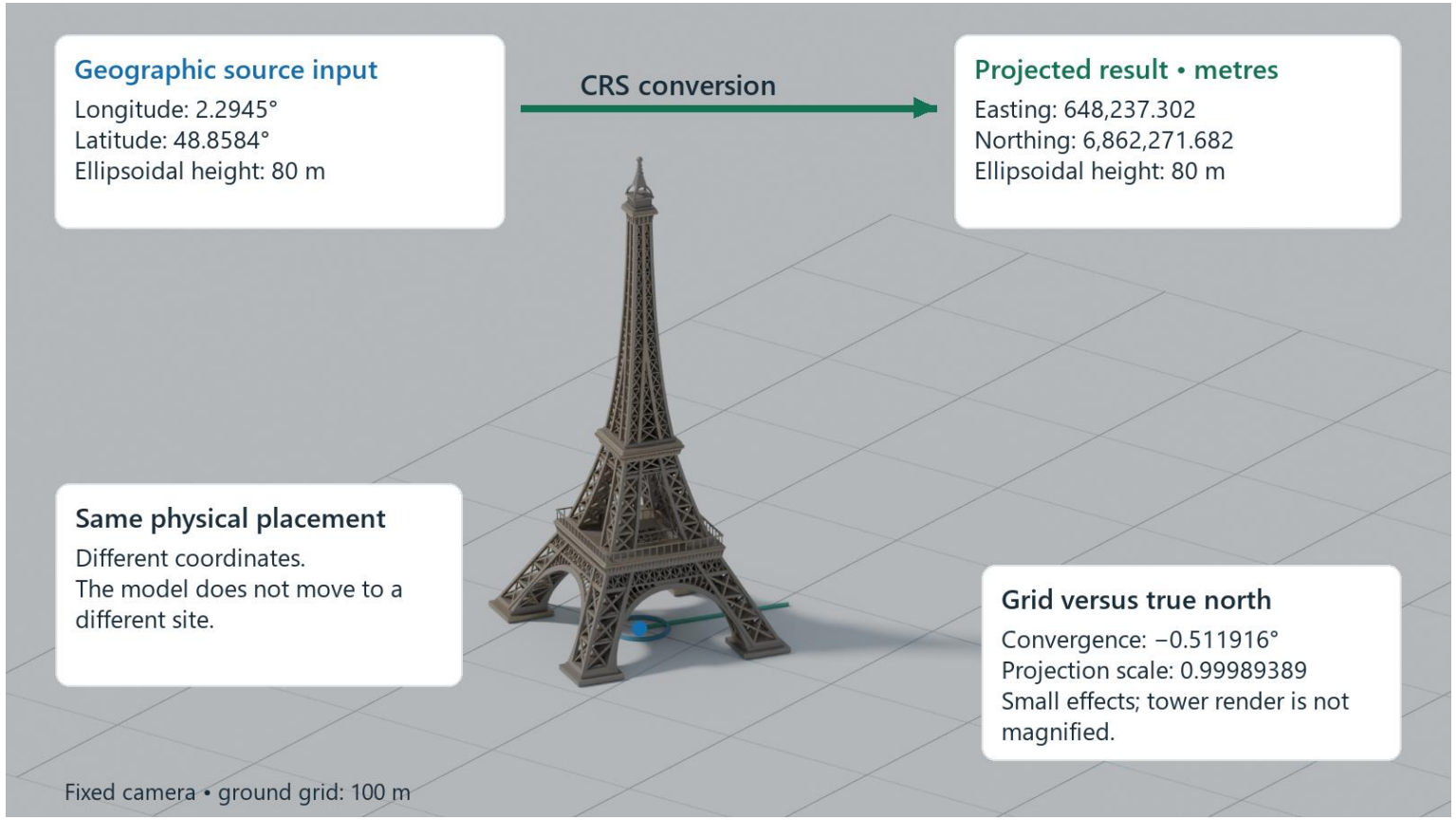
Can a reader distinguish the model's shape from its location on Earth?

Source information	Plain meaning
Model geometry	Shape and local distances, with declared units and up direction. In USD: points, metersPerUnit and upAxis.
Coordinate reference	A CRS binding refers to WKT, the standard text description of the location's axes, units and height meaning.
Geographic origin	crs:position locates the model origin. The illustrative example uses longitude 2.2945°, latitude 48.8584° and ellipsoidal height 80 m.
Physical orientation	crs:orientation records heading, pitch and roll in degrees: which way the model faces and tilts against local east, north and up.

A reader can place the model without treating longitude or latitude as a distance or adding computed results to the source schema.

One physical place, different coordinates

Does changing coordinate systems preserve physical placement?



Informative illustration. Limits in the notes and README.

Observed

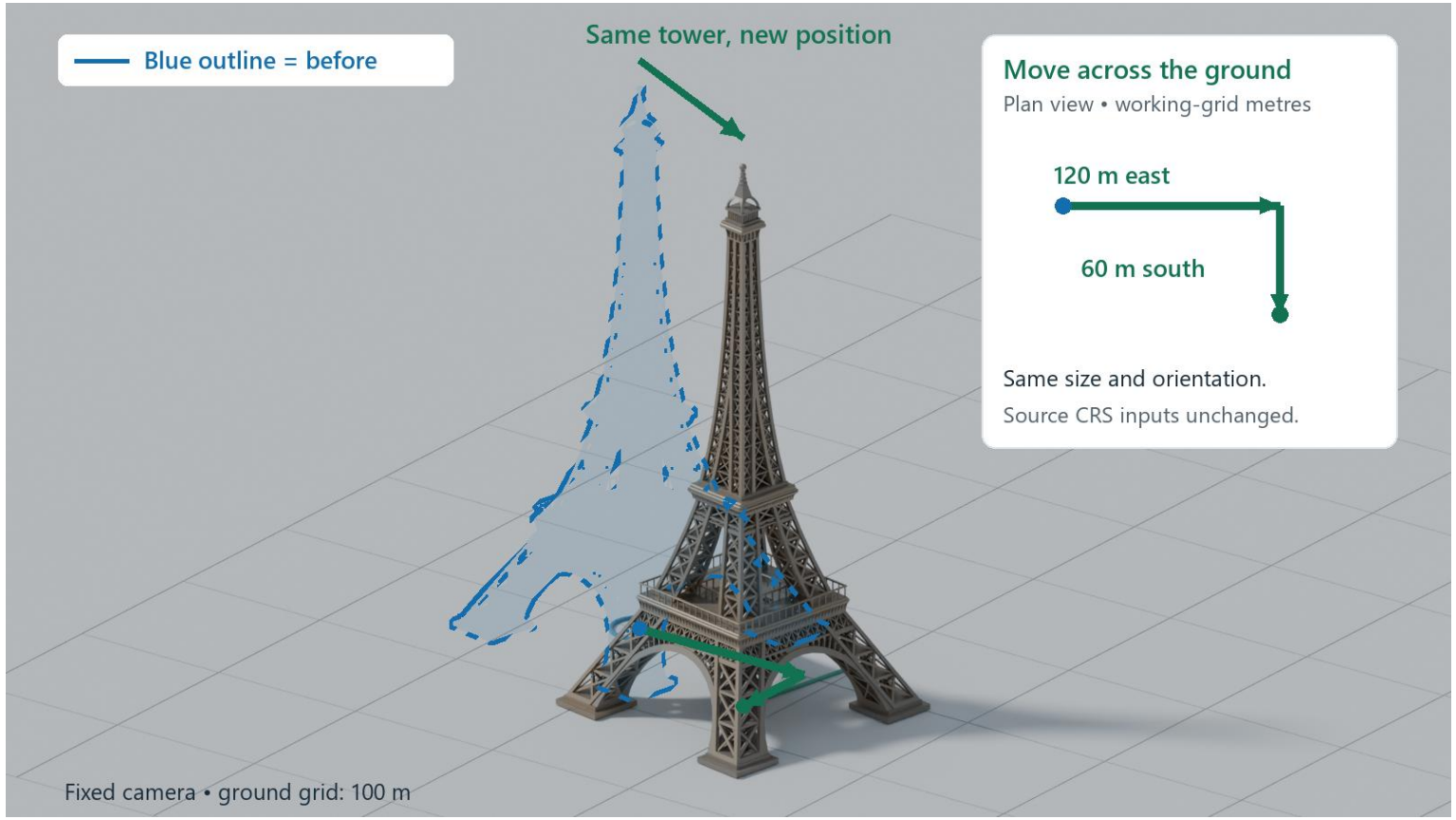
Longitude, latitude and height become map coordinates. The physical site stays the same, while grid north and local scale can differ.

What it establishes

A new coordinate representation has a defined physical meaning. The France and Colorado numerical controls test conversion separately from this illustration.

Ordinary model adjustments after placement

In which coordinates do an author's scale, rotation and translation operate?



Informative illustration. Limits in the notes and README.

Observed

The illustration scales by 1.15, rotates 20° and moves 120 m east and 60 m south. It shows adjustments on the directly bound model.

What it establishes

The draft distinguishes project adjustments from model-local child offsets. A numerical control separates those interpretations by 4.222 m.

The same rules in two placement readers

Can separate implementations derive the same coordinates from the proposed inputs and rules?

France: CC49 to Lambert-93, metres

Resolved component	Python reader	C++ reader
Map east coordinate	661,101.109417	661,101.109417
Map north coordinate	6,857,834.256536	6,857,834.256536
Height	37.298000	37.298000

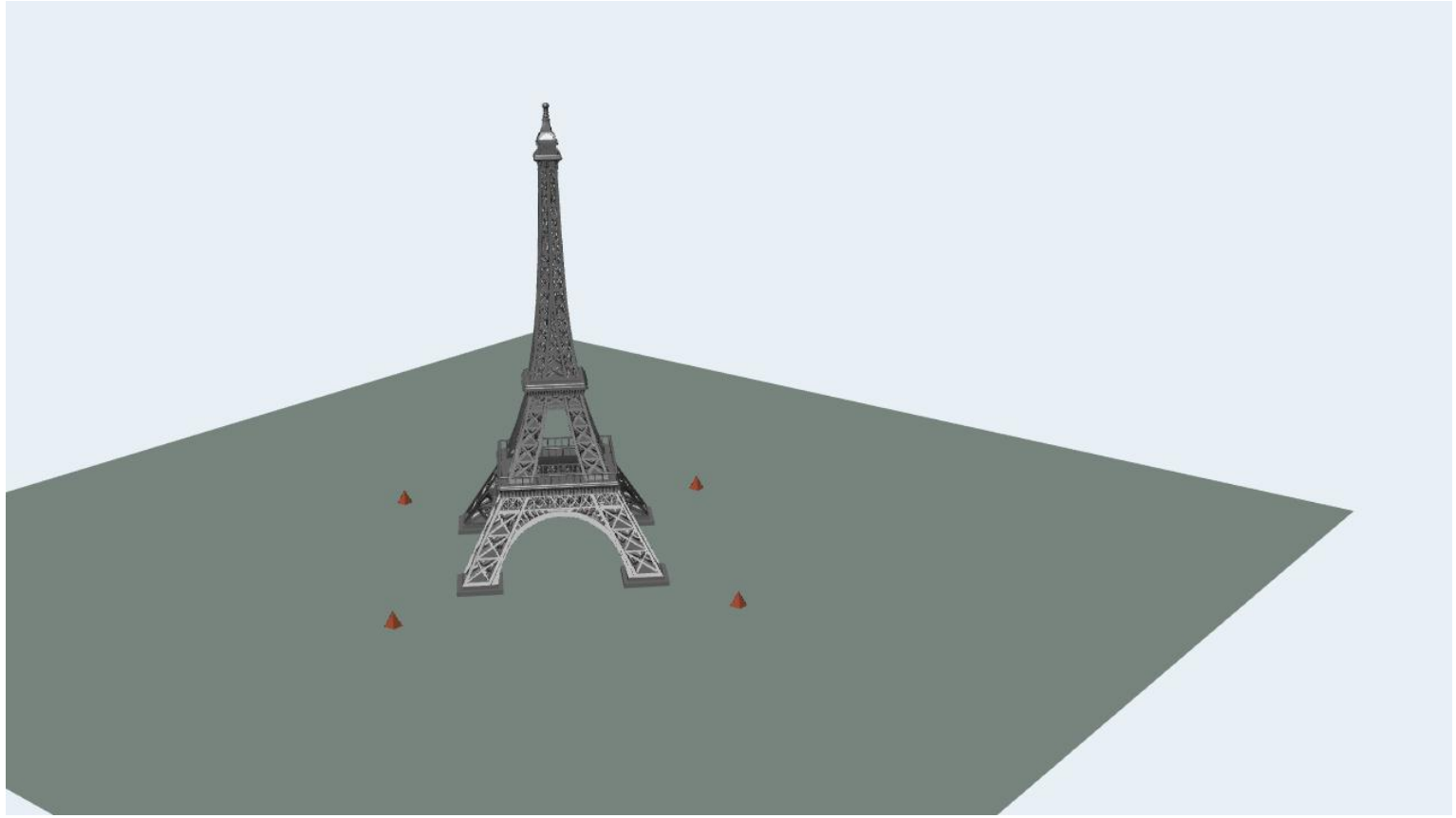
Colorado: State Plane to site grid, US survey feet

Resolved component	Python reader	C++ reader
Map east coordinate	3,108,871.450101	3,108,871.450101
Map north coordinate	1,206,317.469403	1,206,317.469403
Height	5,484.677949	5,484.677949

Both readers use PROJ and share native dataset decoding.

Ordinary USD geometry for rendering

Can a renderer use the result without implementing geospatial evaluation?



Recorded native render. Limits in the notes and README.

Observed

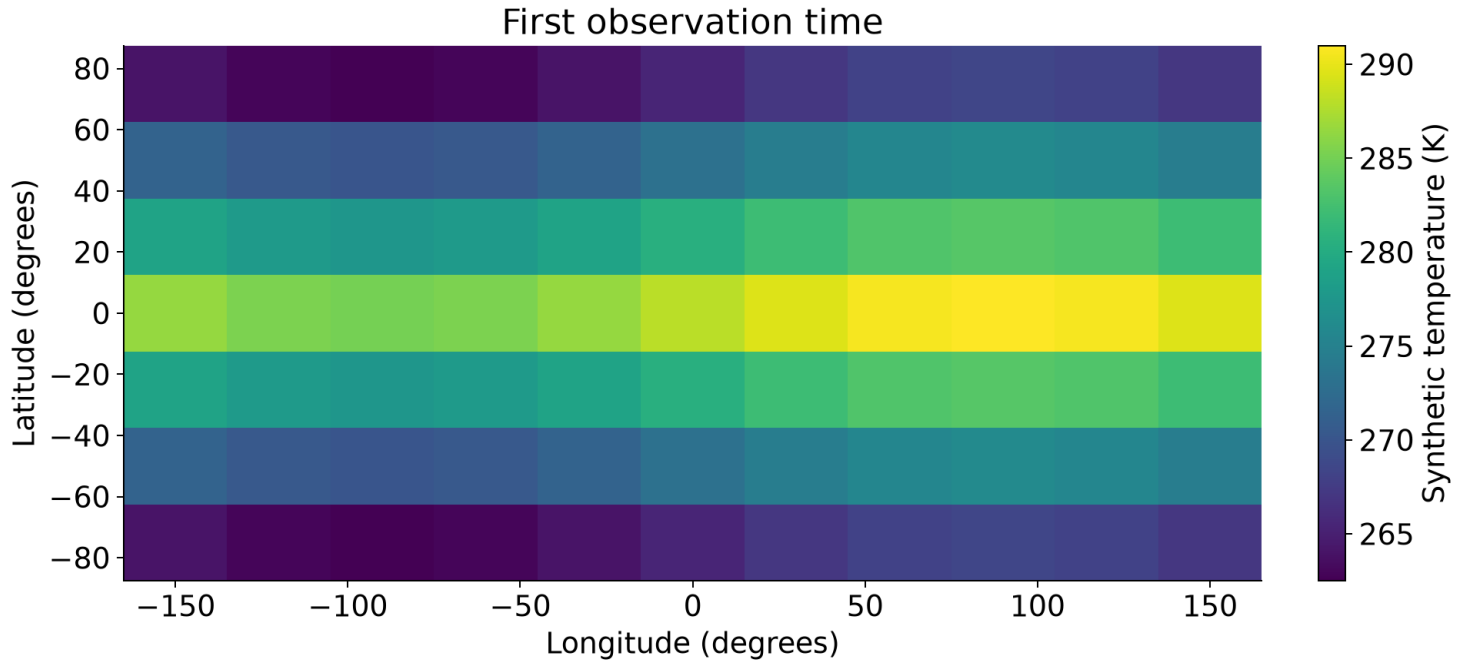
Native Hydra/Storm renders the C++-resolved Eiffel bake. A double-precision origin keeps local geometry small.

What it establishes

The visualization path uses ordinary USD geometry. Reassembly controls confirm that a parent translation and an authored reset keep their USD meaning.

Measurements stay paired with their locations

Can a native measurement dataset remain usable without rewriting values or guessing coordinate meaning?



The global field returns geographic and Earth-centred positions for 154 samples, preserving native values and two observation times.

Synthetic temperatures in kelvin. Both readers share decoding, so their agreement alone cannot verify the association.

Processing requirements before loading content

Can a consumer discover required geospatial processing while the content payload stays closed?

Closed-payload summary	Consumer meaning
Complete-positive	Declared hard dependency: geospatial processing required
Complete-empty	Publisher-qualified known-none: no retained geospatial use
Unknown	Missing qualification supplies no proof of no dependency

The three summary cases retain closed payloads during detection. A separate publisher check catches a stale empty claim that retains a georeferenced variant.

Selected payload and retained-variant controls. Fixed v1 names and the standards approval owner still need group agreement.

WKT writing, reading and comparison

Can different WKT spellings remain usable without losing coordinate meaning?

Writers emit lossless normal text. Readers accept supported valid non-normal WKT with a warning. Normalized text identity and semantic CRS equivalence remain separate.

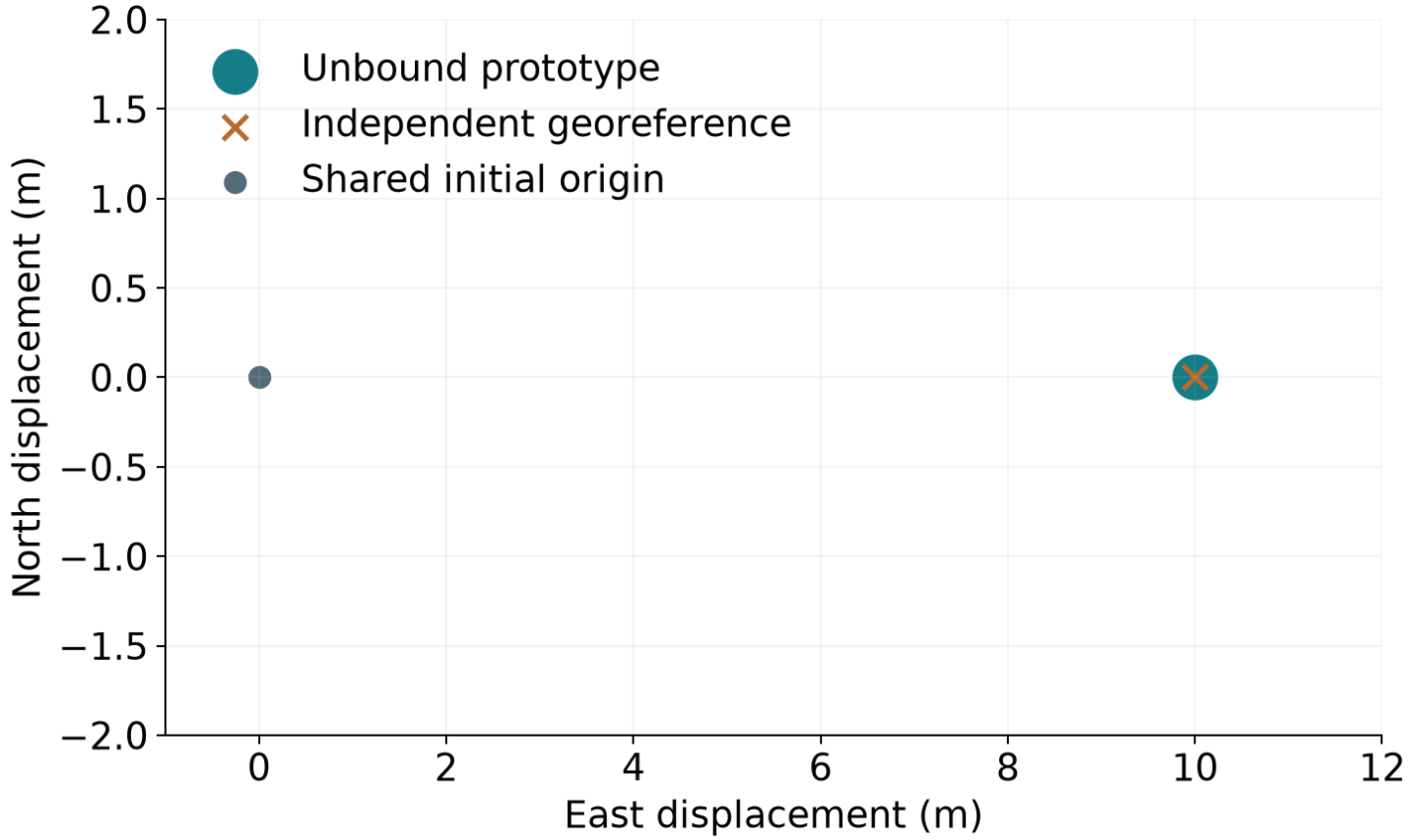
Text before writing	Writer normal form
1.00	1
1E0	1
0.50	0.5

Both placement readers resolve each spelling to the expected coordinates. The Omniverse stage writeback checks the same cases. A quoted keyword remains ordinary name text.

This lexical profile remains proposed. Invalid syntax and unsupported coordinate meaning still fail. Coordinate epochs remain deferred.

Repeated models retain their placement

Does an independently georeferenced prototype receive an instance offset exactly once?



Plot of recorded synthetic results. Limits in the notes and README.

Observed

Both readers return Earth-centred displacement [0, 10, 0] m for the two prototypes. Rotation and scale controls also distinguish the frame choices.

What it establishes

The selected controls test the occurrence contract rather than merely the prototype origin. An unavailable required inverse fails visibly.

Proposal quality and its limits

Selected demonstrations support the proposed behavior.

Review dimension	Assessment
Data model and runtime rules	The draft separates authored inputs from computed results and defines placement, adjustment, query and bake behavior.
Group alignment	Input and declaration rules have agreement. Detailed conventions and identities await review.
Prototype limitations	Lighting directions lack an error bound. Rotation uncertainty is unquantified.
Verification coverage	Selected workflows and distinguishing controls support the rules. Broader composition and continuous space/time coverage remain unverified.

No additional undefined model rule was confirmed in the inspected paths. That is a bounded finding.

Appendix: A construction site's own survey grid

Can a project's local survey grid participate as a defined CRS?

Recorded native render. Limits in the notes and README.

Observed

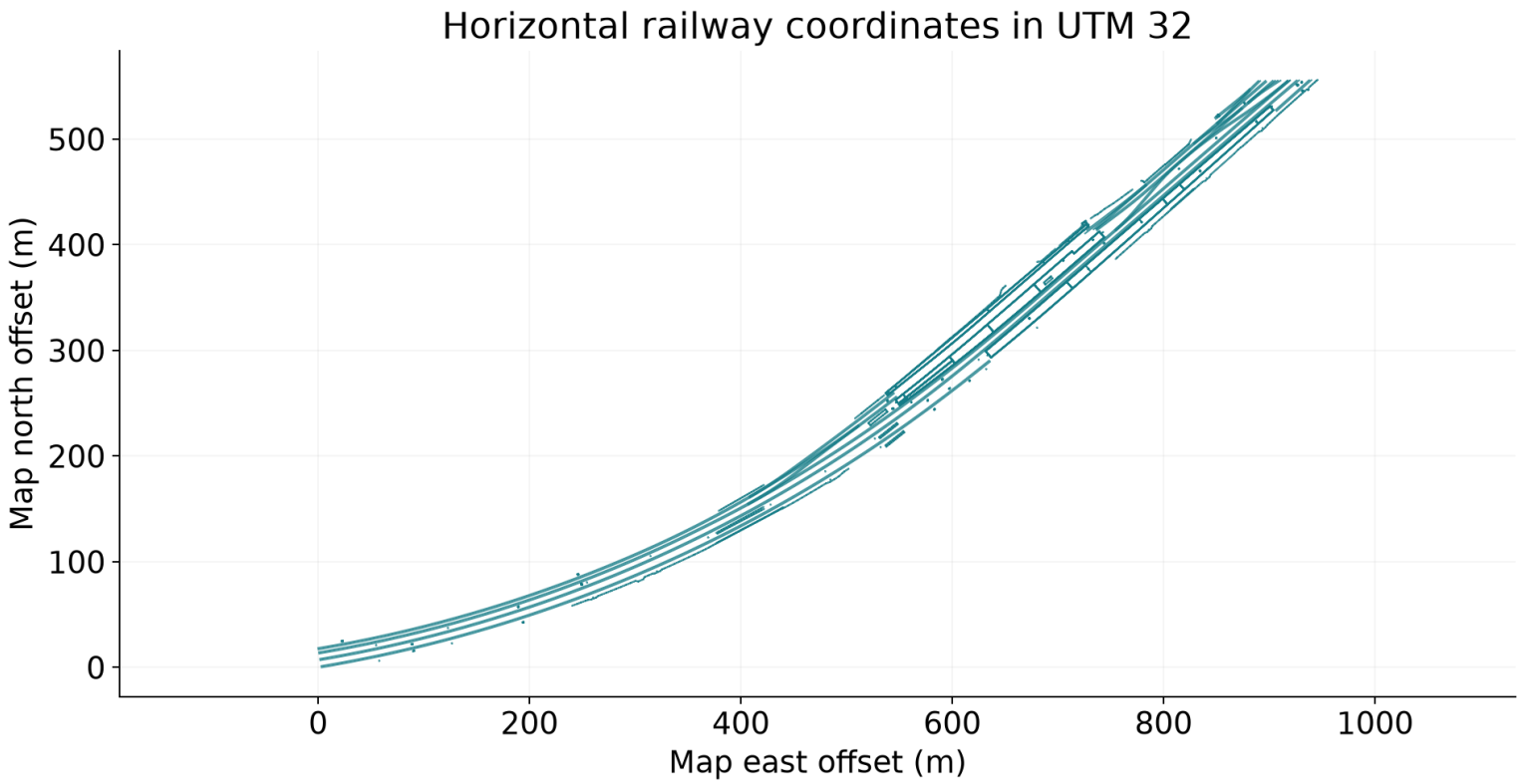
The original LandXML survey breaklines resolve and reach an ordinary USD bake rendered by Storm.

What it establishes

The partner case exercises a site's calibrated CRS. Its WKT-carried calibration participates in resolution.

Appendix: Horizontal railway placement

Can incomplete source metadata still support a useful, explicitly horizontal workflow?



Plot of recorded resolved geometry. Limits in the notes and README.

Observed

The horizontal copy resolves 15,822 vertices across 2,386 parts. The plot uses offsets only for display.

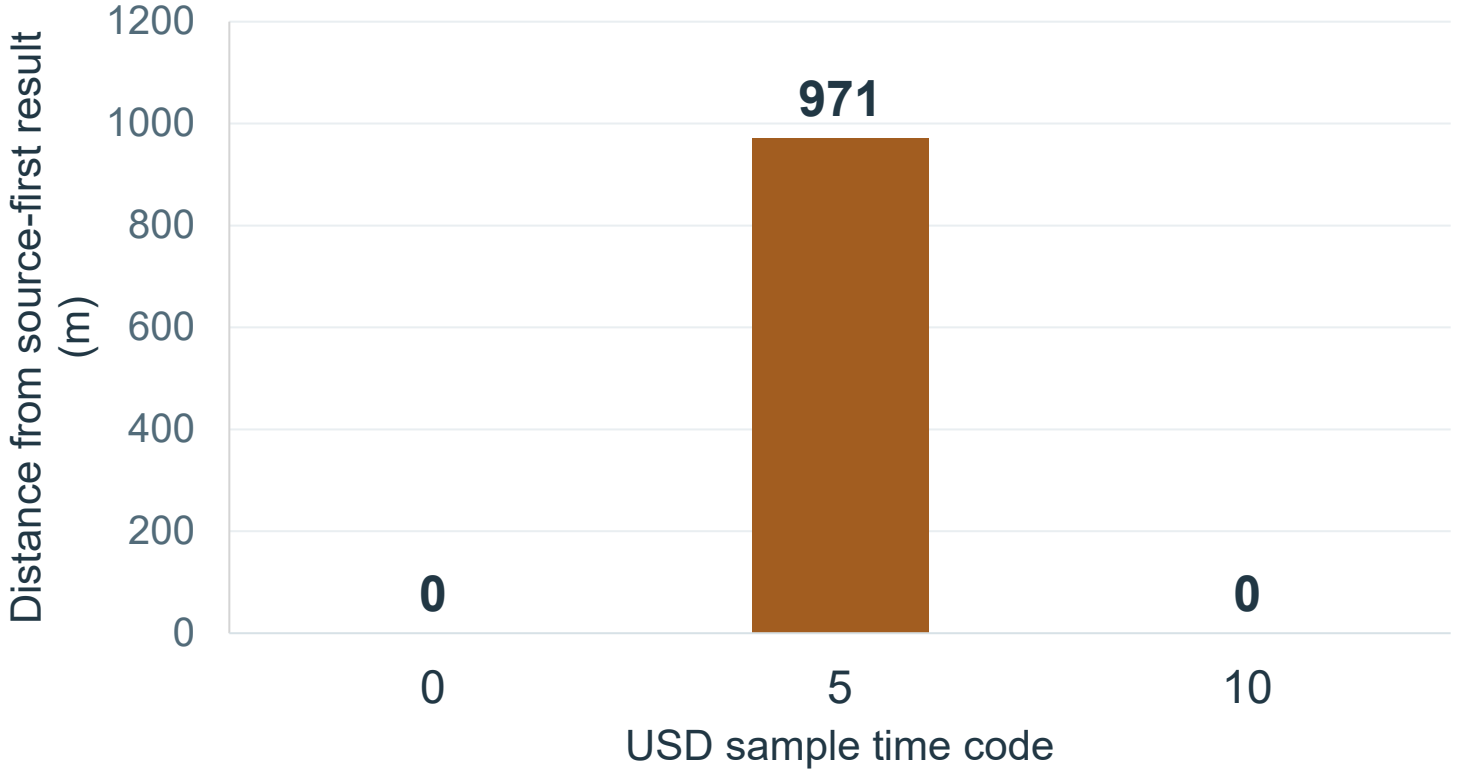
What it establishes

The draft supports horizontal resolution without inventing a height reference. The original third components remain uninterpreted.

Appendix: Animation evaluation order

Interpolate the authored source position first, then convert coordinates.

Equatorial longitude changes from 0° to 2°. Interpolating converted endpoints misses the source-first midpoint by 971.421 m.



Why the order matters

The order is testable. Sample keys alone cannot certify the original trajectory between samples.

The bake preserves time codes 0, 5 and 10 at 48 time codes per second. Continuous trajectory coverage remains unverified.

Appendix: Placement conventions

Topic	Proposed meaning to review
Physical orientation	One heading/pitch/roll tuple describes physical attitude against local east, north and up. Projection effects stay computed.
Project adjustments	The directly bound model's own xformOps adjust its placed result in a fixed project chart, including ordinary pivots.
Child objects	Child transforms define model-local geometry. Their offsets follow model axes and the geospatial attitude.
Repeated models	Each instance offset retains its instancer-local meaning. Independent prototype placement participates once. A required unavailable inverse fails.
Geographic scene results	Geographic queries retain longitude/latitude/height. Cartesian geometry and bounds use the datum's Earth-centred frame.

These are proposed definitions in the full draft.

Appendix: Data and declaration conventions

Topic	Proposed meaning to review
Native data associations	Declare the asset, field and coordinate domain. Preserve named dimensions and observation times.
WKT string normalization	Normalize lexical spelling without losing represented information. Equal tokens establish serialized identity.
Semantic CRS comparison	Different normalized texts can represent equivalent CRSs. Comparison names its criteria and domain.
Dependency declaration	Profiles records complete publication coverage, including unloaded and retained variant content. Fixed v1 identities remain proposed.
Baked coordinate context	Retain a Cartesian CRS and double-precision origin without an active source placement binding. Ordinary USD controls subsequent transforms.

Coordinate epochs and scene-authored transformation-resource controls are deferred while preserving a path to future support.

Appendix: Evidence and its limits

Path or check	Recorded evidence
Python and C++ readers	213,263 coordinate results per reader. Maximum measured difference 3.39e-08 m. Shared PROJ and decoding.
Omniverse stage geometry	Reuses Python placement. Readback covers 177,872 vertices and 48 edit, failure and recovery checks across 44 jobs.
Hydra / Storm	Two selected ordinary USD bakes reach native rendering using C++-resolved placements.
Recorded tests	124 contract tests and 70 distinguishing controls pass. Separate diagnostics expose the reported prototype limits.
Source preservation	Original inputs and 50 frozen execution files remain unchanged. The immutable receipt identifies exact executed content.
Bake readbacks	11 selected exports checked. No general material/primvar, physics or continuous trajectory conformance is claimed.

There are two placement readers. Omniverse reuses Python with a verified stage geometry sink. Hydra consumes a C++-resolved ordinary USD bake.